

Zerui Chen | 陈泽瑞

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Graduating Jul 2027 · seeking research engineer / fellowship roles from summer 2027

Education

University of Cambridge | BA (Hons) Computer Science | Cambridge, UK Expected Jul 2027

- **Key Coursework:** Artificial Intelligence, Advanced Data Science, Bayesian Inference, Deep Neural Networks, Natural Language Processing, Geometric Deep Learning, Reinforcement Learning (expected by graduation)
- **Year 2 Grading: First Class** - course rank **18 / 129**

Hwa Chong Institution | GCE A-Levels: 4 H2 AAAA/AA, H3 **Distinction** | Singapore Jan 2016 - Dec 2021

Publications & Research

Search Pathology Diagnosis for Automated Discovery Oct 2026

Co-first author | *NeurIPS 2026 workshops (posters): New in ML · Interpreting Agent Behavior (IAB)*

- Built a **system-agnostic** toolkit to detect **2 types of search pathologies** in automated discovery systems (ADS), based on a unified intermediate representation of traces and a bias-robust protocol to tune and evaluate the detectors.
- Demonstrated the detectors through a case study on a **SOTA ADS**, yielding **28% improvement in sampling efficiency**.

Evolution Fine-Tuning: Learning to Discover Across 371 Optimization Tasks May 2026

Co-author | *NeurIPS 2026 (poster) · Oral, AID-Wild Workshop @ ACM CAIS 2026 · arXiv:2606.29082*

- Contributed **nanodiscover** as the **test-time-RL scaffold** behind the paper's test-time-RL results.

nanodiscover 2026

Co-author | *Open-source research software*

- A clean reimplementation of **TTT-Discover**, a **SOTA test-time-learning automated discovery system**, replacing its Tinker API dependency ($\approx \$500$ per problem) with local vLLM/DeepSpeed/Ray inference and training. Fixes upstream bugs, reproduces the published results up to sampling stochasticity, and was adopted by Evolution Fine-Tuning.

Comparative study of decision trees vs deep learning models for atrial tachycardia diagnosis Jun 2022

First author | *arXiv:2206.07515*

Professional Experience

WorldQuant | Quantitative Research Intern | Singapore Jul 2026 - Sep 2026

- Designed and built a **tiered forecast-evaluation suite** for residualized return forecasts: **headline metrics** spanning accuracy, calibration and a **PnL proxy**, complemented by **diagnostics** for calibration shape, tail effects and realisation dynamics, with **robust inference** throughout via HAC standard errors and simultaneous confidence bands.
- Merged into prod. Now focused on improving modelling accuracy with feature engineering and augmented objectives.

ByteDance, Global E-Commerce | Machine Learning Engineer Intern | Singapore Jul 2025 - Sep 2025

- Researched and shipped novel and robust **auxiliary objectives** and **model architectural** optimizations for TikTok Global Shop's fine-ranking DL model, trained on in-house **HPC** clusters, achieving **+0.3% revenue lift**.
- Built internal tools for interactive model architecture **visualization** and automated **training behavior analysis**.

[Competition] NUS Medical Grand Challenge | Full-Stack & LLM Developer | Global Feb 2024 - Aug 2024

- Founded medical-AI startup *Tomy* for remote stoma care and emerged **Grand Champion** amongst 40 teams from universities across Asia, partnering with **5 Singapore public hospitals** with deployed clinical trials.
- Implemented and deployed image classification API using few-shot prompted vision language model, achieving **100% recall** and **80% precision** in 4-fold cross-validation with extremely scarce training data.

Singapore Armed Forces | Signals & Admin Specialist | Singapore Jan 2022 - Nov 2023

- Developed duty assignment algorithm with availability constraints, saving hours monthly and improving fairness.

Agency for Science, Technology and Research | Bio-AI Research Intern | Singapore Jun 2020 - May 2021

- Developed a LSTM-CNN deep learning model for atrial tachycardia diagnosis, reaching SOTA-comparable 81% accuracy while being highly data-efficient; parallelized data ETL pipeline to achieve **100× throughput**.

Technical Skills

- **Machine Learning** with Python using PyTorch, vLLM, DeepSpeed, Ray, LightGBM for LLM inference and fine-tuning and deep learning for NLP, CV, RecSys, with Lightning, Slurm and AWS for experiment orchestration.
- **Data ETL & Analysis** with SQL, MongoDB, NumPy, Pandas, Matplotlib
- **General Development** with C++, Java, OCaml, Rust, TypeScript, Swift
- **Languages:** English (native), Mandarin Chinese (native).